

COMPSCI 1DM3: Discrete Mathematics for Computer Science

Course Outline – Winter 2026

McMaster University

This course outline contains important information that may affect your grade. You should retain it and refer to it throughout the semester, as you will be assumed to be familiar with the rules specified in this document.

Instructor

Angela Zavaleta Bernuy

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Web: <https://angelazb.github.io>

Office Hours: By appointment.

Schedule

Lectures	Mon, Wed, Thu	4:30 PM - 5:20 PM	Angela Zavaleta Bernuy	TSH B128
Tutorial-5	Tue	2:30 PM - 4:20 PM	Sohaib Ahmed	ETB 124
Tutorial-6	Mon	8:30 AM - 10:20 AM	Eric Lee	PGCLL M22
Tutorial-7	Tue	2:30 PM - 4:20 PM	Stephen Tang Tang	HH 102
Tutorial-8	Tue	2:30 PM - 4:20 PM	Eden Ofek Nalian	BSB 121

The lecture and tutorial schedule is based upon current University and public health guidelines and may be subject to changes during the term. Any changes to the schedule or course delivery will be communicated on the course announcements section on Avenue to Learn. Please check the announcements prior to attending class.

Course Web Site

This course will be administered via Avenue to Learn. Go to <http://avenue.mcmaster.ca/> to access the course's Avenue to Learn page. Please send only normal email to the instructional staff; do not send mail via Avenue.

Students should be aware that, when they access the electronic components of this course, private information such as first and last names, user names for the McMaster email accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in this course will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the Instructor. *It is the student's responsibility to be aware of the information on the course's Avenue to Learn page and to check regularly for announcements.*

Required Resources

1. *Course web site:* All course materials will be available on Avenue to Learn (Avenue) at <http://avenue.mcmaster.ca/>.
2. *Textbook:* Kenneth H. Rosen, *Discrete Mathematics and Its Applications*, McGraw Hill, 2019.

Major Topics

1. Logic and proofs
2. Basic structures: Sets, functions, sequences, sums, matrices
3. Algorithms
4. Induction and recursion
5. Number theory and modular arithmetic
6. Counting
7. Discrete probability
8. Graphs and trees

Sections of Textbook to be Covered

- Chapter 1: Sections 1.1 - 1.8 (Review)
- Chapter 2: Sections 2.1 - 2.4, 2.6 (Review)
- Chapter 3: Sections 3.1 - 3.3
- Chapter 4: Sections 4.1 - 4.3
- Chapter 5: Sections 5.1 - 5.4
- Chapter 6: Sections 6.1 - 6.4
- Chapter 7: Sections 7.1 - 7.4
- Chapter 10: Sections 10.1 - 10.4 (time permitting)
- Chapter 11: Sections 11.1, 11.2, (time permitting)

Depending on the time, the instructor might need to change this list during the course.

Work Plan

There will be lectures, tutorials, assignments, a discussion forum in Avenue, two midterm tests, and a final exam. Students are strongly encouraged to read sections of the textbook assigned for each lecture *before* coming to the class and actively engage in the discussion during the lecture. Students are expected to participate in the discussion forum and tutorials by asking questions. **Each student is expected to devote a minimum of ten hours of work per week for this course.** There will be recitation notes and extra examples that students can use to engage in the tutorials.

Tutorials will be led by a TA during which students can ask for clarifications of a subject (from the lecture or the recitation notes) or for hints for problems (either from extra examples or the assignments). Students are expected to participate in a discussion forum either on Avenue (under Communication / Discussions) or Teams. Students are encouraged to use the forum to discuss the course material and to ask and answer questions. The Instructor and TAs will monitor the forum and participate in the discussion as needed.

Please note the following:

- **Assignments:** There will be five assignments consisting of problems requiring mathematical constructions and traditional-style proofs. Each assignment will be posted with clear due date. Take the deadline very seriously! **Late submission policy:** You will incur a penalty of 10% for each hour the assignment is submitted late.
- There will be one midterm test accounting for 25% of the final grade. The confirmed date and location will be announced as soon as possible.
- The final exam, which is cumulative, will be 2.5 hours long. It will take place on the date scheduled by the University.
- Both the Midterm and the Final are closed-book (which means you may not refer to any notes or books during the exam period). Also, no formula sheet is allowed.

Evaluation

The course grade will be based on the student's performance on assignments, midterm tests, and the final exam as follows:

Assignments (5)	25%
Quizzes (5)	10%
Midterm	25%
Final exam	40%
Total	100%

The Instructor reserves the right to adjust the marks for an assignment, midterm test, or final exam by increasing or decreasing every score by a fixed number of points.

Other Policy Statements

1. Significant study and reading outside of class is required.
2. Students are expected to engage during lecture sessions and tutorials by asking questions.
3. If you plan on getting an MSAF, inform the instructor immediately. For assignments an MSAF will give you a 3-day extension. If a student submits an MSAF for a quiz or the midterm test, the weight will be added to the final exam.
4. If there is a problem with the marking of an assignment, the student should first discuss the problem with the TA who marked it. An assignment mark will only be changed if the problem is reported within two weeks of the date that the assignment mark was posted on Avenue.
5. No electronic devices (calculators, cell phones, etc.) may be used during midterm tests and the final exam.
6. Emails with a source address outside of McMaster University will not be read by the instructional staff.

Communication Policy

Most of students' questions regarding the assignments will be answered by TAs via solving related examples during tutorials. Therefore, we strongly recommend attending all the tutorial sessions. To communicate with course TAs reliably, students should Email the TA member using the email listed under TA team members. The instructor/TAs may not be available for immediate replies to your communication outside of class hours. Permit up to a business day for response before following up on urgent issues, and up to two business days for non-urgent issues.

The instructor/TAs may not answer questions related to the assignment 24 hours before each submission deadline. Students are expected to either join the tutorials or ask questions before the latter mentioned constraint.

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/), located at <https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/>.

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity/Plagiarism Detection

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Use of Generative AI

Students are not permitted to use generative AI in this course. In alignment with [McMaster academic integrity policy](#), it "shall be an offence knowingly to submit academic work for assessment

that was purchased or acquired from another source”. This includes work created by generative AI tools. Also state in the policy is the following, “Contract Cheating is the act of “outsourcing of student work to third parties” (Lancaster & Clarke, 2016, p. 639) with or without payment.” Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

Courses With An On-Line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Academic Accommodation Of Students With Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s [Academic Accommodation of Students with Disabilities](#) policy.

Requests For Relief For Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).¹

Academic Accommodation For Religious, Indigenous Or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright And Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.

¹Updated to the Policy on Requests for Relief for Missed Academic Term Work effective May 1, 2021